

Windows Endpoint Security Monitoring

Splunk and Sysmon home-lab incident investigation

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Environment: Isolated VirtualBox lab

Classification: Portfolio / Educational

Executive summary

I built an isolated Windows monitoring lab and used it to investigate a controlled sequence of authentication, process, account, privilege, and persistence events. Windows security logs and Sysmon telemetry were forwarded to Splunk Enterprise, where I created searches, a dashboard, and two scheduled alerts.

The investigation identified five failed SMB logons against victim.user from 192.168.56.102, PowerShell launched with ExecutionPolicy Bypass, creation of svc_backup, addition of an account to the local Administrators group, and creation of the scheduled task Windows Update Check Service. Both scheduled alerts were validated successfully.

Outcome: telemetry collection, investigation logic, visualization, and alert execution were all validated end to end.

Purpose and scope

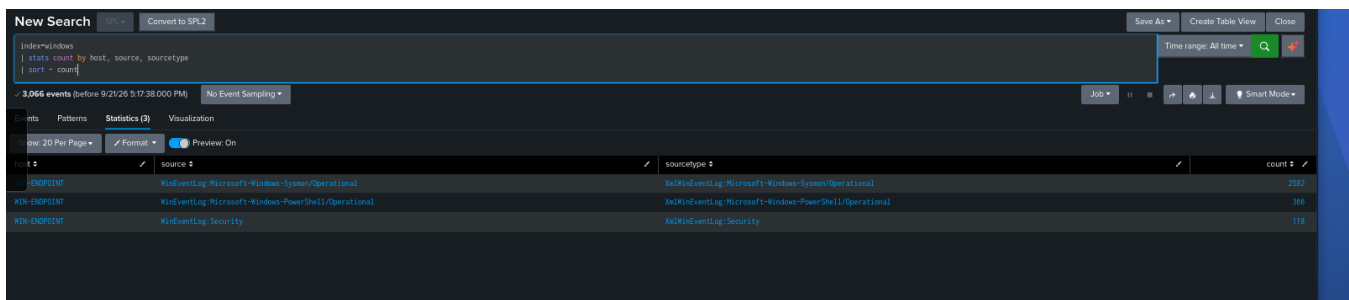
The purpose was to practice an entry-level SOC workflow rather than emulate destructive malware. All commands were executed on systems I own and control. The lab contains one monitored endpoint and one analyst/SIEM host, so conclusions are limited to this environment.

Lab architecture and collection

System	Address / role	Function
Kali Linux	192.168.56.102	Splunk Enterprise, analyst workstation, and source of SMB tests
WIN-ENDPOINT	192.168.56.103	Windows 11 endpoint monitored with Security logs and Sysmon
Universal Forwarder	TCP 9997	Forwards Windows telemetry to index=windows

Collection validation

I verified the Windows audit policy, installed Sysmon with a configuration file, confirmed that the SplunkForwarder service was running, tested TCP connectivity to port 9997, and confirmed three Windows sources in Splunk: Security, PowerShell Operational, and Sysmon Operational.



The screenshot shows the Splunk search interface with the following details:

- Search Bar:** Contains the query `index=windows | stats count by host, source, sourcetype | sort - count`. The time range is set to "All time".
- Results:** 3,066 events (before 9/21/25 5:17:38.000 PM). The results are displayed in a table with columns: host, source, sourcetype, and count.
- Table Data:**

host	source	sourcetype	count
WIN-ENDPOINT	WinEventlog:Microsoft-Windows-Sysmon/Operational	KaliWinEventlog:Microsoft-Windows-Sysmon/Operational	202
WIN-ENDPOINT	WinEventlog:Microsoft-Windows-PowerShell/Operational	KaliWinEventlog:Microsoft-Windows-PowerShell/Operational	166
WIN-ENDPOINT	WinEventlog:Security	KaliWinEventlog:Security	118

Figure 1. Windows event sources arriving in the Splunk windows index.

Data handling

Events arrived as XML. I used SPL regular expressions to extract fields such as EventID, TargetUserName, IpAddress, Image, CommandLine, MemberName, and TaskName. These fields became the basis for correlation and alert logic.

Incident timeline

Time (EDT)	Event	Observation
17:26:15	4624	Successful logon for victim.user
17:27:31	Sysmon 1	PowerShell with ExecutionPolicy Bypass and reconnaissance commands
17:29:46	4720	Local account svc_backup created
17:29:46	4732	Account SID added to local Administrators
17:29:48	4698	Windows Update Check Service scheduled task created
17:48:46-17:49:04	4625	Five failed logons from 192.168.56.102 against victim.user
18:46:52 and 18:49:21	Sysmon 1	Two controlled PowerShell alert-validation executions

The complete machine-readable timeline is included in data/incident_timeline.csv and in the evidence workbook.

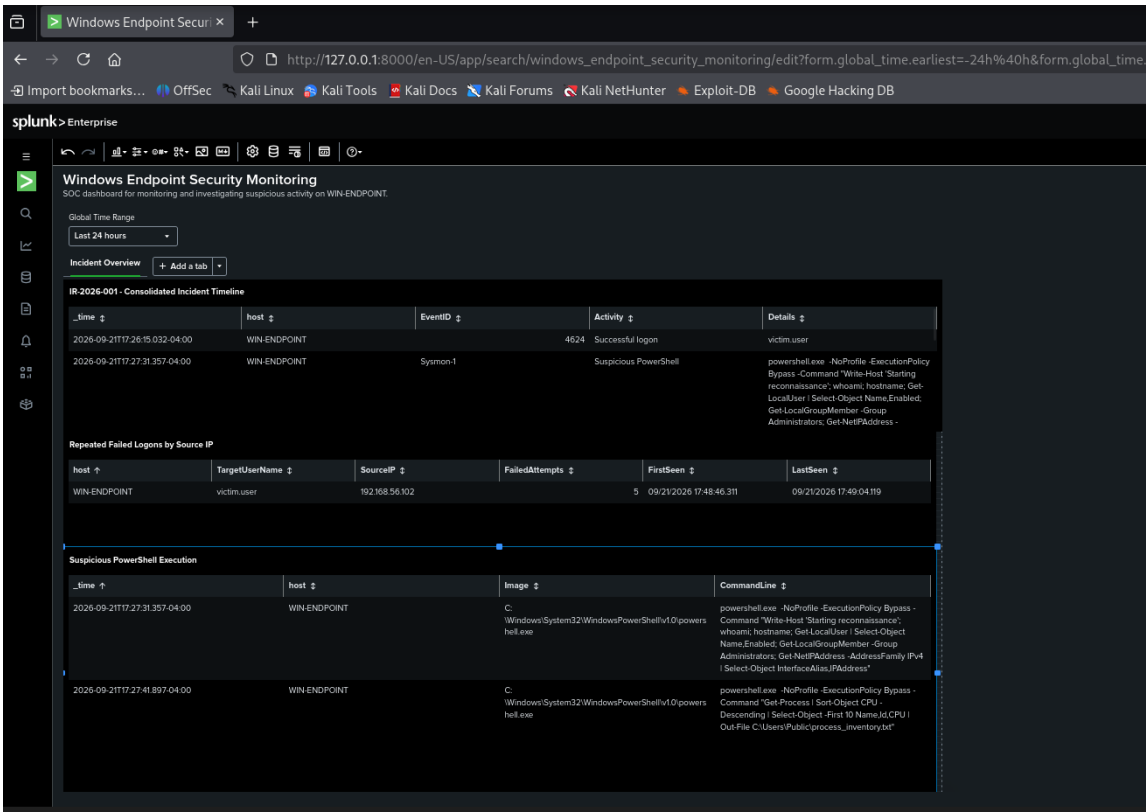


Figure 2. Final dashboard with the incident timeline, failed-logon summary, and PowerShell results.

Findings and analysis

Finding	Assessment	Evidence
Repeated failed logons	Five Event ID 4625 records shared the same target user and source IP. This met the lab threshold and is consistent with password guessing.	192.168.56.102 -> victim.user
Suspicious PowerShell	Sysmon captured powershell.exe with ExecutionPolicy Bypass. The argument is not malicious by itself, but it warrants review with the full command line and user context.	Sysmon Event ID 1
Account creation	svc_backup was created shortly after the reconnaissance-style activity.	Security Event ID 4720
Privileged group change	A local SID was added to Administrators, increasing account privileges.	Security Event ID 4732
Scheduled task	Windows Update Check Service was created to run PowerShell at logon as SYSTEM.	Security Event ID 4698

Analyst judgment

No single event proves compromise. The security value comes from the sequence: authentication, PowerShell, account creation, privilege modification, and scheduled-task creation in a short period. In a real environment I would confirm whether each action was approved, review the parent process and user session, examine adjacent network activity, and isolate the endpoint if the activity were unauthorized.

MITRE ATT&CK; mapping

Behavior	Technique
Password guessing	T1110.001
PowerShell	T1059.001
Local account	T1136.001
Account manipulation	T1098
Scheduled task	T1053.005

Detection logic and validation

Detection 1: repeated failed logons

The search filters Security events to Event ID 4625, extracts the target account and source IP, groups by host, user, and source, and requires at least five events. It also records the first and last timestamps. The scheduled alert runs every five minutes and searches the previous ten minutes.

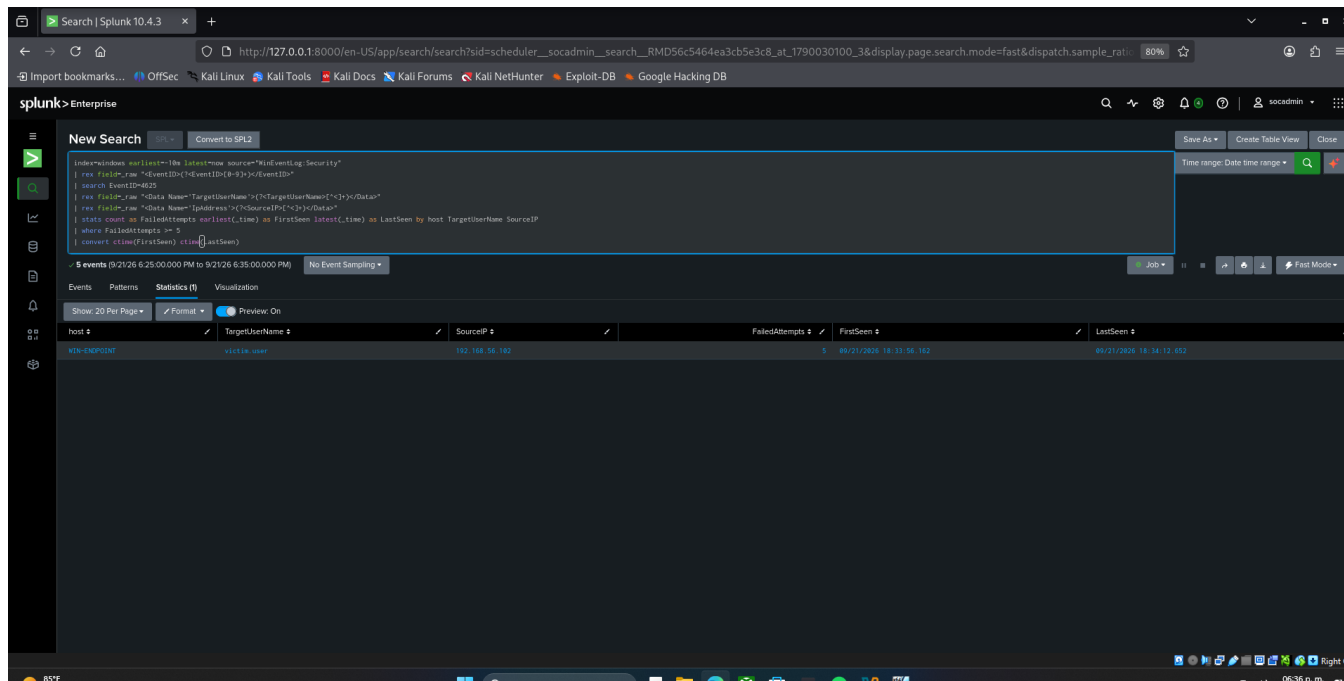


Figure 3. Alert result showing five failed attempts from 192.168.56.102 against victim.user.

Detection 2: suspicious PowerShell

The search filters Sysmon Event ID 1 and checks whether CommandLine contains ExecutionPolicy Bypass. I executed a benign validation command twice. Splunk returned two matching process-creation events, demonstrating that the rule and collection pipeline worked.

PowerShell alert evidence

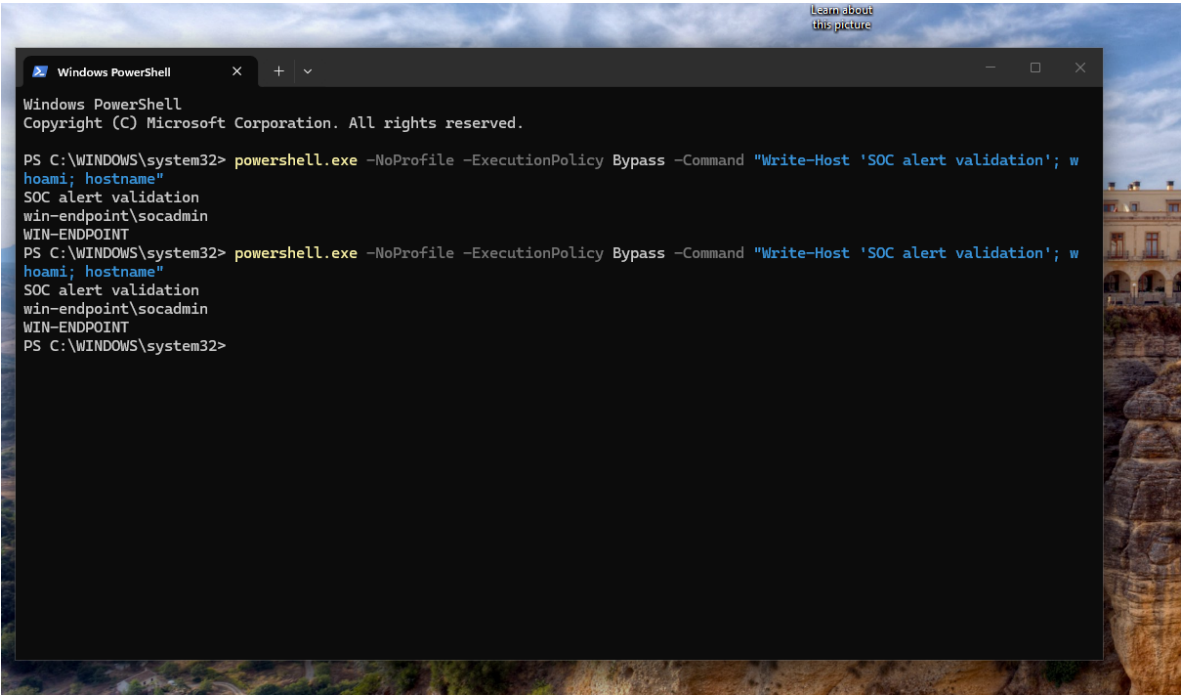


Figure 4. Two benign PowerShell validation executions on WIN-ENDPOINT.

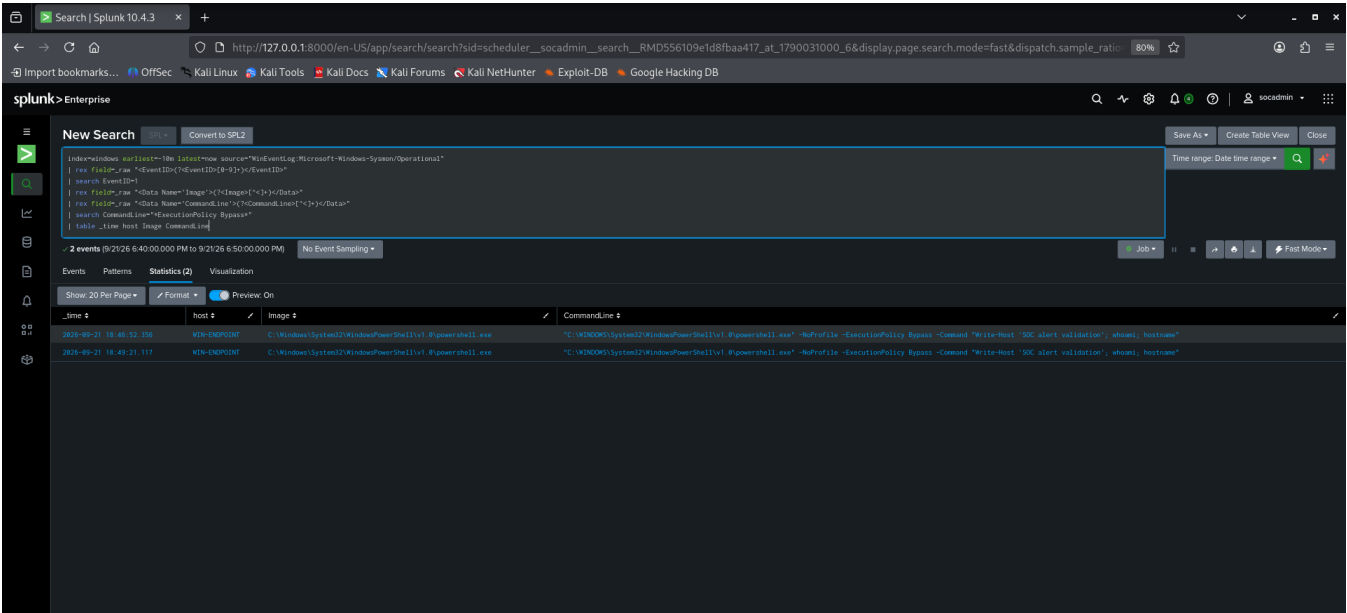


Figure 5. Two Sysmon events returned by the scheduled PowerShell detection.

Response recommendations

Priority	Recommended action
Immediate	Confirm whether the user, PowerShell command, account creation, group change, and scheduled task were authorized.
Immediate	If unauthorized, isolate the endpoint, disable affected accounts, preserve logs, and collect volatile evidence.
Short term	Add approved administrator accounts, management tools, and maintenance windows to the detection context.
Short term	Enable alert suppression or throttling to control duplicate detections from overlapping windows.
Long term	Add endpoint, identity, DNS, firewall, and asset-criticality enrichment; test response playbooks regularly.

False-positive considerations

ExecutionPolicy Bypass is commonly used by legitimate automation and software deployment tools. Failed logons can result from a mistyped password, stale credentials, or a service account. Scheduled tasks and local account changes can also be legitimate administration. Production alerting should consider process ancestry, signer information, user role, asset sensitivity, frequency, and approved change records.

Lessons learned

The largest lesson was that detection engineering is an end-to-end process. A correct SPL query is not enough if audit policy, forwarding, time windows, or alert scheduling are wrong. Troubleshooting network reachability and field extraction was part of the work, not separate from it. I also learned to validate a rule with controlled activity and retain both the action and the SIEM result as evidence.

Limitations

This report covers one Windows endpoint and synthetic events in an isolated lab. It does not establish production baselines or measure alert precision over time. The supplied rules are starting points for tuning, not universal indicators of compromise.

Evidence index

File	Purpose
01-splunk-enterprise-running.png	Platform running
02-windows-audit-policy.png	Audit policy
03-sysmon-installation.png	Sysmon installation
04-forwarder-connectivity.png	Forwarder and port 9997
05-windows-data-ingestion.png	Log ingestion
06-scheduled-task-persistence.png	Scheduled task simulation
07-failed-logon-test.png	Five SMB failures
08-failed-logons-detected.png	Failed-logon search
09-powershell-detected.png	PowerShell search
10-account-and-task-events.png	Account and task events
11-soc-dashboard.png	Final dashboard
12-failed-logon-alert-triggered.png	Trigger history
13-failed-logon-alert-results.png	Failed-logon result
14-powershell-validation-command.png	Validation command
15-powershell-alert-results.png	PowerShell alert result

Supporting files

The repository also includes the SPL searches, normalized timeline CSV, indicators CSV, Spanish learning guide, and evidence workbook.